Test Layer 3 Cross-Connect Traffic: test_l3.py

2025-07-23-12-53-17

Objective

The Layer 3 Traffic Generation Test is designed to test the performance of the Access Point by running layer 3 Cross-Connect Traffic. Layer-3 Cross-Connects represent a stream of data flowing through the system under test. A Cross-Connect (CX) is composed of two Endpoints, each of which is associated with a particular Port (physical or virtual interface).

Device Under Test Information

Device Under Test	DUT Name	
	DUT Hardware Version	
	DUT Software Version	
	DUT Serial Number	

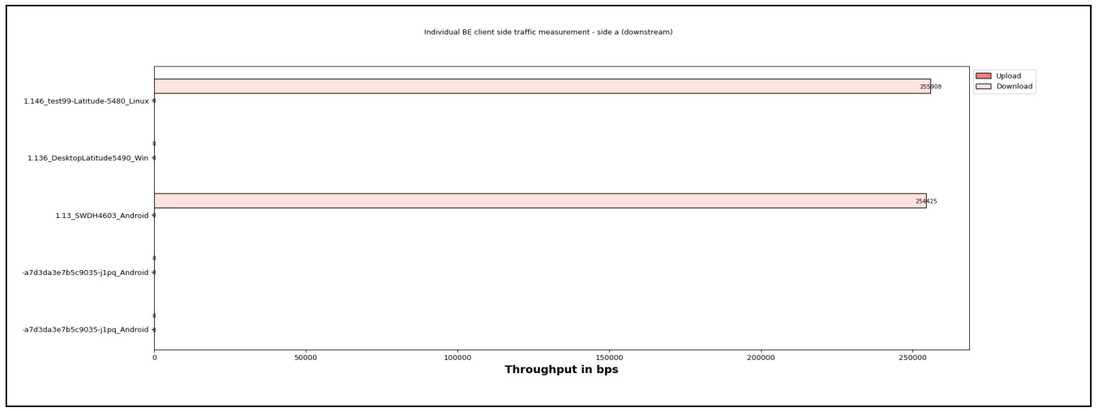
Test Configuration

Test Configuration	LANforge ip	192.168.244.97
	LANforge port	8080
	Upstream	eth1
	Test Duration	3m
	Polling Interval	5s
	Total No. of Devices	5

Radio Configuration

Individual throughput mcast download bps: 256000 traffic BE (WiFi).

The below graph represents individual throughput for 5 clients running BE (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".

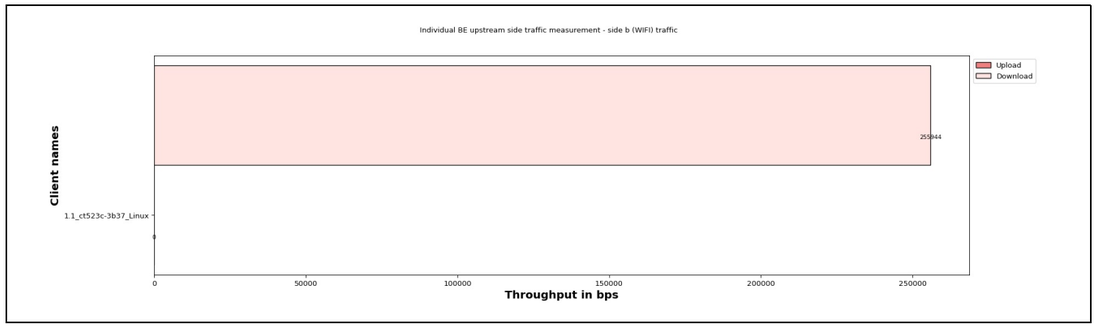


Client Alias	Host eid	Host Name	Device Type / Hw Ver	Endp Name	Port Name	Mode	Mac	SSID	Channel	Type of traffic	Traffic Protocol	Offered Upload Rate Per Client	Offered Download Rate Per Client	Upload Rate Per Client	Download Rate Per Client	Drop Percentage (%)
1.11_r-a7d3da3e7b5c9035-j1pq_Android	1.11	r-a7d3da3e7b5c9035-j1pq	Google Pixel 4a r13 sdk: 33	MLT-mrx-BE-wlan0-1	1.11.wlan0	802.11abgn-20	3a:3b:b5:f3:1c:28	xiab_2G_WPA2	6	BE	Mcast	0	256000	0	0	0.0
				MLT-												

1.11_r-a7d3da3e7b5c9035-j1pq_Android	1.11	r-a7d3da3e7b5c9035-j1pq	Google Pixel 4a r13 sdk: 33	mrx-BE-wlan1-2	1.11.wlan1	802.11abgn 20	02:00:00:00:00:00	xiab_2G_WPA2	6	BE	Mcast	0	256000	0	0	0.0
1.13_SWDH4603_Android	1.13	SWDH4603	samsung SM-J600G r10 sdk: 29	MLT-mrx-BE-wlan0-3	1.13.wlan0	AUTO 20	fe:80:53:eb:57:18	xiab_2G_WPA2	6	BE	Mcast	0	256000	0	254425	0.0
1.136_DesktopLatitude5490_Win	1.136	DesktopLatitude5490	Win/x86 6.2	MLT-mrx-BE-wlan0-4	1.136.wlan0	802.11abgn 20 1x1	dc:8b:28:28:d7:64	xiab_2G_WPA2	6	BE	Mcast	0	256000	0	0	0.0
1.146_test99-Latitude-5480_Linux	1.146	test99-Latitude-5480	Linux/x86-64	MLT-mrx-BE-wlan0-5	1.146.wlan0	802.11bgn 40 2x2	00:24:d6:1a:20:f1	xiab_2G_WPA2	6	BE	Mcast	0	256000	0	255908	0.0

Individual throughput upstream endp, offered upload bps: 0 offered download bps: 256000 /station for traffic BE (WiFi).

The below graph represents individual throughput for 1 clients running BE (WiFi) traffic. Y- axis shows "Client names" and X-axis shows "Throughput in Mbps".



Client Alias	Host eid	Host Name	Device Type / HW Ver	Endp Name	Port Name	Mode	Mac	SSID	Channel	Type of traffic	Traffic Protocol	Offered Upload Rate Per Client	Offered Download Rate Per Client	Upload Rate Per Client	Download Rate Per Client	Drop Percentage (%)
1.1_ct523c-3b37_Linux	1.1	ct523c-3b37	Linux/x86-64	MLT-mtx-BE-eth1-0	1.1.eth1		00:60:e0:88:3b:38		[channel: 1]	BE	Mcast	0	256000	0	255944	0.0

Layer 3 Cx Traffic Last Reporting Interval eth1

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent
1753255718	07_23_2025_12_58_38	10	0	0	256000	256000	MTU	MTU	MTU	MTU	-1	eth1	215311	4396246	1 Gbps	1 Gbps	NaN	NaN	NaN	00:60:e0:88:3b:38	[channel: 1]	0	0	0	0	0	0	0	0	0	0

Layer 3 Cx Traffic Last Reporting Interval 1.11.wlan0

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent
1753255718	07_23_2025_12_58_38	10	0	0	256000	256000	MTU	MTU	MTU	MTU	-1	1.11.wlan0	1035789	110847	96 Mbps	96 Mbps	-26	7E:D2:94:48:2C:A9	802.11abgn 20	3a:3b:b5:f3:1c:28	6	0	0	0	0	0	0.0	0	0	0	0

Layer 3 Cx Traffic Last Reporting Interval 1.11.wlan1

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent

1753255718	07_23_2025_12_58_38	10	0	0	256000	256000	MTU	MTU	MTU	MTU	-1	1.11.wlan1	0	0	96 Mbps	96 Mbps	-26	7E:D2:94:48:2C:A9	802.11abgn 20	02:00:00:00:00:00	6	67	0	0	0	0	0.0	0	0	0	0

Layer 3 Cx Traffic Last Reporting Interval 1.13.wlan0

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent
1753255718	07_23_2025_12_58_38	10	0	0	256000	256000	MTU	MTU	MTU	MTU	-1	1.13.wlan0	1298248	86703	72 Mbps	72 Mbps	-25	7E:D2:94:48:2C:A9	AUTO 20	fe:80:53:eb:57:18	6	-1556	0	254393	263255	3996	0.0	0	0	0	0

Layer 3 Cx Traffic Last Reporting Interval 1.136.wlan0

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent
1753255718	07_23_2025_12_58_38	10	0	0	256000	256000	MTU	MTU	MTU	MTU	-1	1.136.wlan0	1443350	3375	144.4 Mbps	144 Mbps	-23 dBm	7E:D2:94:48:2C:A9	802.11abgn 20 1x1	dc:8b:28:28:d7:64	6	-3	0	0	0	0	0.0	0	0	0	0

Layer 3 Cx Traffic Last Reporting Interval 1.146.wlan0

Time epoch	Time	Total-Station-Count	UL-Min-Requested	UL-Max-Requested	DL-Min-Requested	DL-Max-Requested	UL-Min-PDU	UL-Max-PDU	DL-Min-PDU	DL-Max-PDU	Attenuation	Name	Rx-Bps	Tx-Bps	Rx-Link-Rate	Tx-Link-Rate	RSSI	AP	Mode	MAC	Channel	Rx-Latency	Rx-Jitter	Ul-Rx-Goodput-bps	Ul-Rx-Rate-ll	Ul-Rx-Pkts-ll	Ul-Rx-Drop_Percent	Dl-Rx-Goodput-bps	Dl-Rx-Rate-ll	Dl-Rx-Pkts-ll	DL-Rx-Drop-Percent
1753255718	07_23_2025_12_58_38	10	0	0	256000	256000	MTU	MTU	MTU	MTU	-1	1.146.wlan0	1292203	2744	144.4 Mbps	144.4 Mbps	-19 dBm	7E:D2:94:48:2C:A9	802.11bgn 40 2x2	00:24:d6:1a:20:f1	6	-6	0	255888	263317	4004	0.0	0	0	0	0